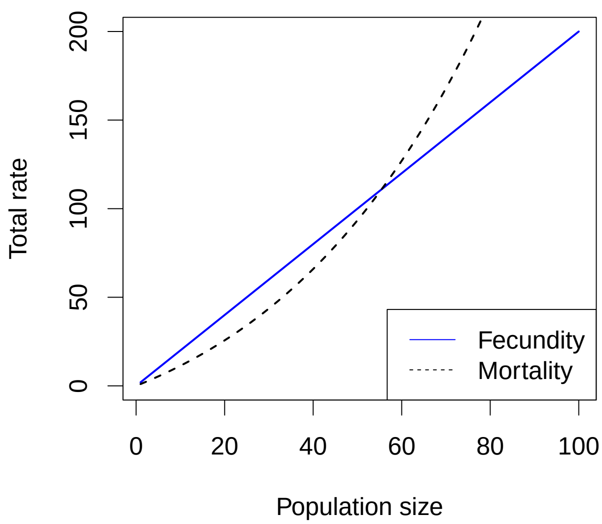
1. The bacteria you are studying enter a resting state, but only when under stress, typically due to crowded conditions. If the resting state has lower mortality than the bacteria normally experience when not stressed, we would also expect that, compared to unstressed conditions, the resting state has ________ birth rates and ________ reproductive number $\mathcal{R}$.

A. **lower; lower**
B. lower; higher
C. higher; lower
D. higher; higher

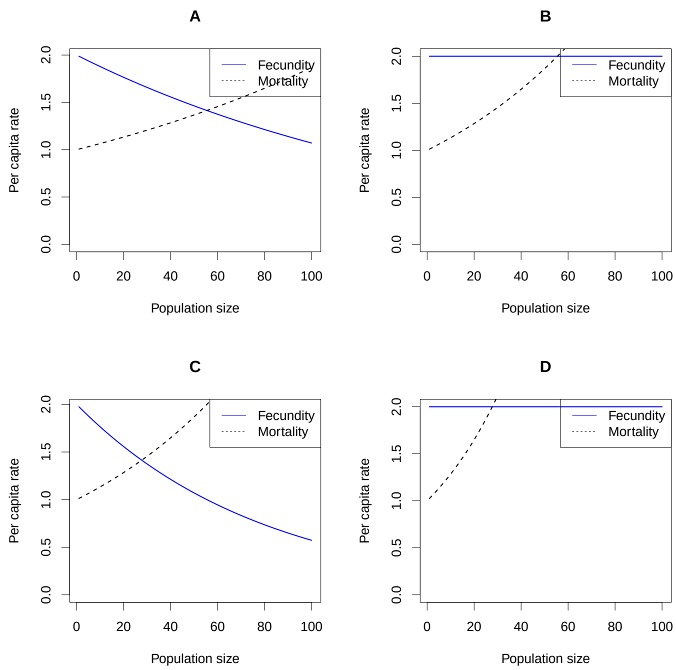


The figure above shows the assumptions made for a discrete-time birth-death model. *Use it for the next 3 questions.*

2. The figure shows:

A. No density dependence
B. Density dependence in fecundity only
C. **Density dependence in mortality only**
D. Density dependence in both mortality and fecundity

3. Which of the four pictures below was generated by the same model as the picture above?



ANS: B

4. A population following this model will:

A. Increase exponentially without limit
B. Decrease exponentially to zero
C. **Approach an intermediate equilibrium**
D. Decrease to zero if started near zero, and increase to an intermediate equilibrium otherwise

5. In simple, discrete-time models of a single species competing for resources, we often see population cycles:

A. In models where competition is contest-like
B. **In models where competition is scramble-like**
C. In models without competition
D. We don't see population cycles in simple discrete-time models

6. Which of the following would be the strongest reason to prefer a *stage-structured* model to an age-structured model?

A. An annual life cycle
B. A longer life cycle with as usually of a predictable time length (like salmon)
C. **Large variation in size of similar-aged individuals (like hemlock trees)**
D. Small variation in size of similar-aged individuals (like storks)

Use the following information for the next two questions. A population of European beetles introduced to control the invasive weed loosestrife increased from 10/ha in 1995 to 200/ha in 2000. You should assume for the purposes of this question that they display perfect exponential growth.

7. What was their instantaneous growth rate r?

A. 0.30/ha
B. 0.30/yr
C. 0.60/ha
D. **0.60/**yr
E. There is not enough information to tell

8. What was their finite rate of increase λ from generation to generation?

A. 1.35
B. 1.35/ha
C. 1.82
D. 1.82/ha
E. **There is not enough information to tell**

9. Which of the following statements is the most accurate?

A. The log scale is generally better for comparing biological quantities
B. The linear scale is generally better for comparing biological quantities
C. Differences between smaller quantities look relatively bigger on a linear scale than on a log scale
D. **Differences between larger quantities look relatively bigger on a linear scale than on a log scale**

10. *per capita* refers to rates that apply

   A. at densities near the carrying capacity
   B. at very low densities
   C. to a whole population
   D. **to each individual**

11. An ecologist models a population by using the equation, $\frac{dN}{dt} = r_{\max}N(1 - N/K)$, the unit of population density (N) is indiv/km^2 and the unit of time (t) is yr. The units of $r_{\max}$ are __________, and the units of K are __________.

   A. yr; indiv.
   B. yr; indiv/km^2.
   C. 1/yr; indiv.
   D. 1/yr**;** indiv/km^2**.**

12. The carrying capacity for an organism in an environment is the density at which crowding __________ the average of __________ to 1:

   A. reduces; the birth rate
   B. increases; the death rate
   C. **reduces; the reproductive number $\mathcal{R}$**
   D. reduces; the instantaneous growth rate r

13. A pile of radioactive material is decaying *continuously* at an instantaneous rate of 1%/minute. After two *hours*, what proportion is left?

   A. A little more than 98%
   B. Exactly 98%
   C. A little less than 98%
   D. **About 30%**
   E. None

14. A population with a "doubling time" of 20 years is __________ with a "characteristic time" of around __________ years.

   A. decreasing; 14
   B. decreasing; 28
   C. increasing; 14
   D. **increasing; 28**

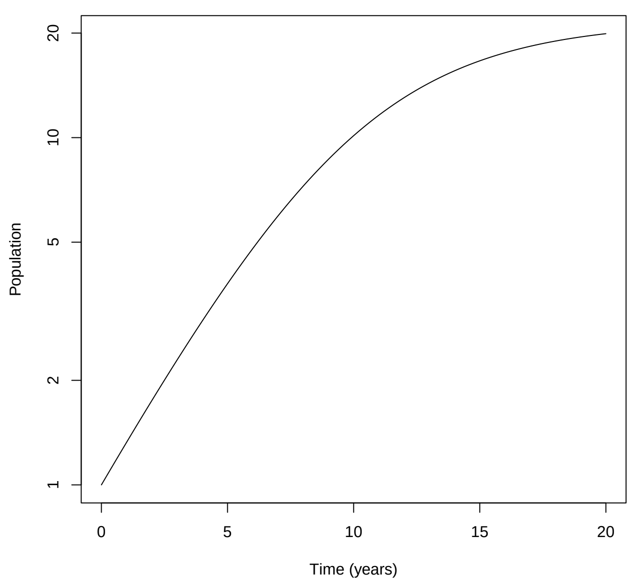
15. A population of whales is estimated to be at stable age distribution, with a constant life table, with reproductive number $\mathcal{R}$= 1.15. It takes the whales several decades to reach maturity and reproduce. This population is

A. declining
B. stable
C. **increasing**
D. showing damped oscillations
E. there is not enough information to answer this question

16. If we are thinking about a simple, continuous-time model, then for a population to be regulated:

A. **The average reproductive number $\mathcal{R}$ must be low at high density and higher at either low or intermediate density**
B. The birth rate b must be low at high density and higher at either low or intermediate density
C. The death rate d must be high at high density and lower at either low or intermediate density
D. All of the above
E. None of the above

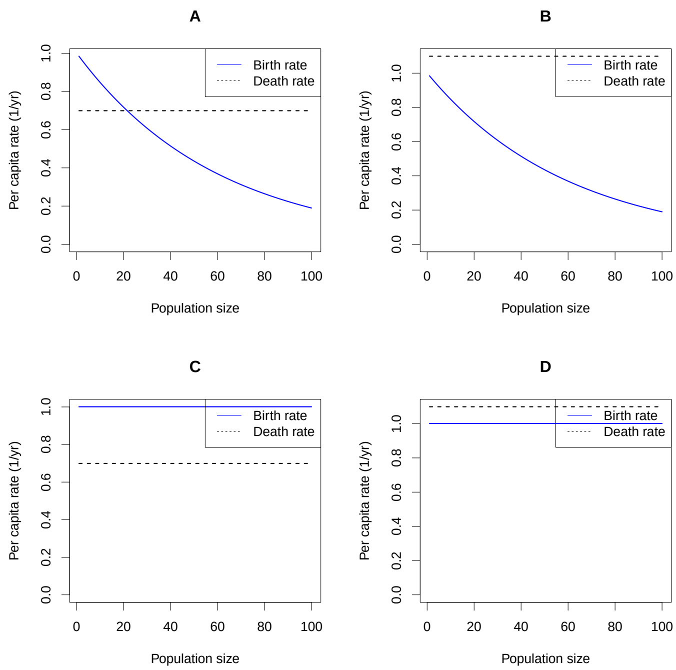
Use the picture below for the next two questions. It shows a time series for a continuous-time birth-death model.



17. This picture shows a population that is:

    A. Increasing arithmetically
    B. Increasing geometrically
    C. **Following a per-capita growth rate with a density-dependent effect**
    D. Following a per-capita growth rate with an Allee effect

18. Which of the four pictures below shows the assumptions that generated this time plot?



ANS: A

19. An exponentially growing rabbit population takes 10 years to grow from 20 individuals to 500 individuals. If it continues to grow exponentially at the same rate, how long would it take to increase from 500 individuals to 2500 individuals?

A. **5 years**
B. 10 years
C. 20 years
D. 40 years
E. 50 years

20. A scientist has made a life table for a perennial population of squirrels, with censusing before reproduction. If she re-does the life table with a different census time (after reproduction, or between seasons), she would expect $\mathcal{R}$ to be __________ because population dynamics look __________ before reproduction.

A. higher; fastest
B. higher; slowest
C. lower; fastest
D. lower; slowest
E. **the same; the same**

21. (2 points) What is the criterion for $\mathcal{R}$ that determines whether a population increases? What about for λ? You may assume these values are stable. What is the difference between $\mathcal{R}$ and λ?

A population expected to increase if $\mathcal{R} > 1$ or if $\lambda > 1$. $\mathcal{R}$ measures the amount by which an individual multiplies itself over a lifetime, and λ measures the amount of multiplication over a time step.

Half a point for each criterion. Half a point each for production or reproduction on each time scale.

22. (3 points). A population of perennial plants has reproductive adults which produce seeds. Some of these seeds survive the winter and sprout. In addition to producing seeds, the adults pass the winter as root systems underground, and some of these resprout in the spring. We estimate the following. Each reproductive plant produces an average of 75 seeds. Each seed has a 0.05 probability of sprouting. Each reproducing adult has a 0.4 probability of resprouting. Each sprout (whether from a seed, or a surviving adult) has a 0.75 probability of surviving to reproduce at the end of the year. We wish to estimate the growth rate using the (unstructured) formula $\lambda = f + p$, so we count just before seed production, when everyone is a reproductive adult. Estimate the value of p, f and λ for this population. Show your work *briefly*.

f is the expected number of new individuals per existing individual: $75\,\mathrm{seed/rA} \times 0.05\,\mathrm{sprout/seed} \times 0.75\,\mathrm{rA/sprout} = 2.812$ (1 point)

Half off for a calculation error. Half off for adding units (since the answer is unitless).

p is the expected probability that an existing individual will be seen again: $0.4\,\mathrm{resprout/rA} \times 0.75\,\mathrm{rA/resprout} = 0.3$ (1 point)

As above

$\lambda = p + f = 3.112$ (1 point).